

General Equilibrium

Ques - 10:- Define consumption externality, Explain by Bandwagon effect and Snob effect.

Answer:-

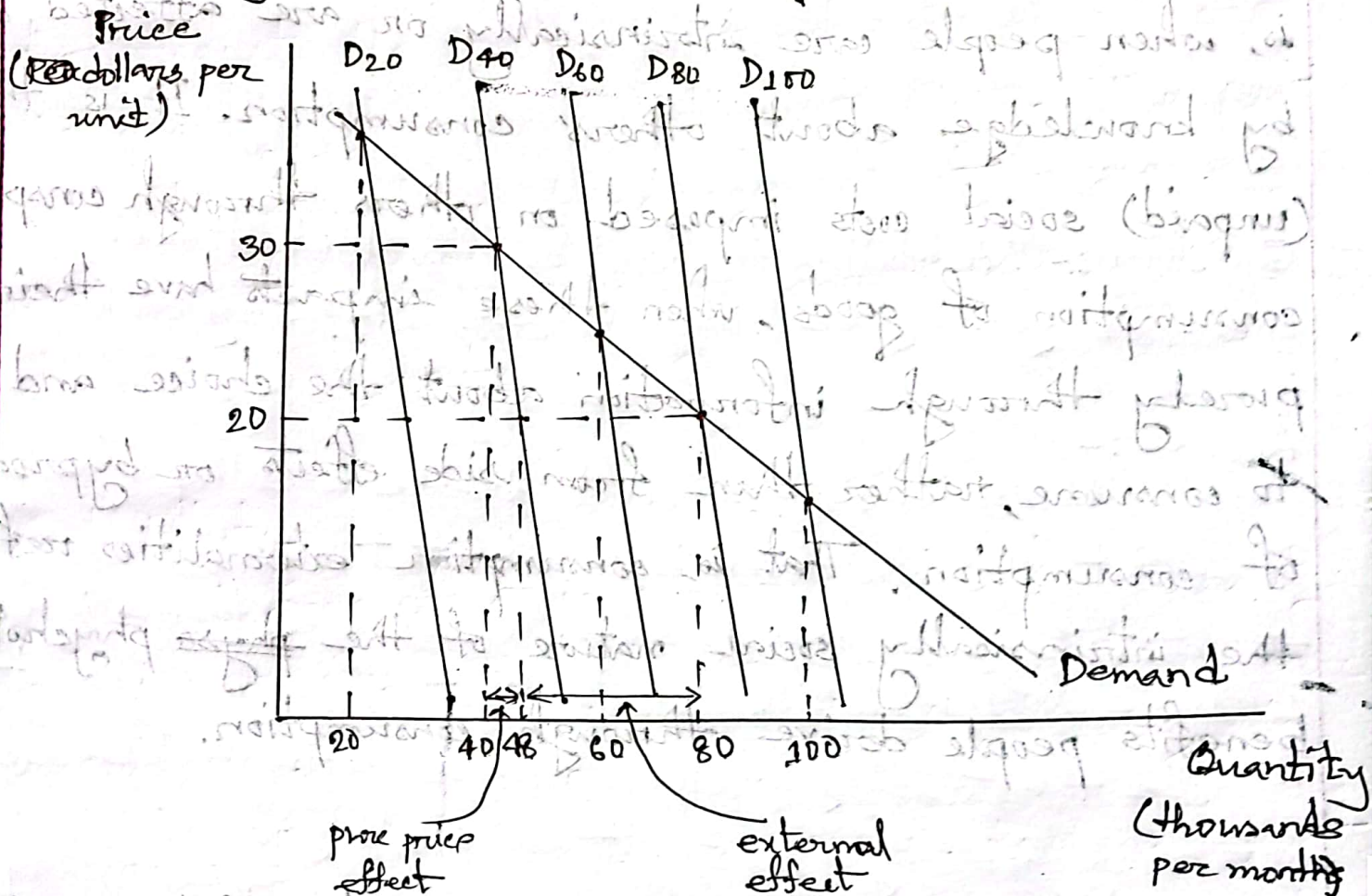
Consumption Externality:-

In economics, consumption externalities exist when the consumption of others matters explicitly and directly in the utility function of individuals - that is, when people care intrinsically or are affected directly by knowledge about others' consumption. It is the (unpaid) social costs imposed on others through conspicuous consumption of goods, when these impacts have their effect purely through information about the choice and ability to consume, rather than from side effects or byproducts of consumption. That is, consumption externalities reflect the intrinsically social nature of the ~~physic~~ psychological benefits people derive through consumption.

Consumption externality can be explained ^{by} two effects:- Bandwagon effect and Snob effect.

Bandwagon Effect:

Bandwagon effect is a positive network externality in which a consumer wishes to possess a good in part because others do. The Bandwagon effect often arises with children's toys (video games, for example). In fact, exploiting this effect is a major objective in marketing and advertising toys. Often it is the key to success in selling clothing.



In the figure, the horizontal axis measures the sales of a product in thousands per month. Suppose consumers think that 20,000 people purchased a certain

product. Because this is a small number relative to the total population, consumers will have little incentive to buy the product. Some consumers may still buy it, but only for its intrinsic value. In this case demand is given by the curve D_{20} .

Suppose instead that consumers think 40,000 people bought the product. Now they find it more attractive and want to buy more. The demand curve is D_{40} . Similarly if consumers think 60,000 people bought it, the demand curve is D_{60} , and so on. The more people consumers believe to have purchased the product, the further to the right the demand curve shifts.

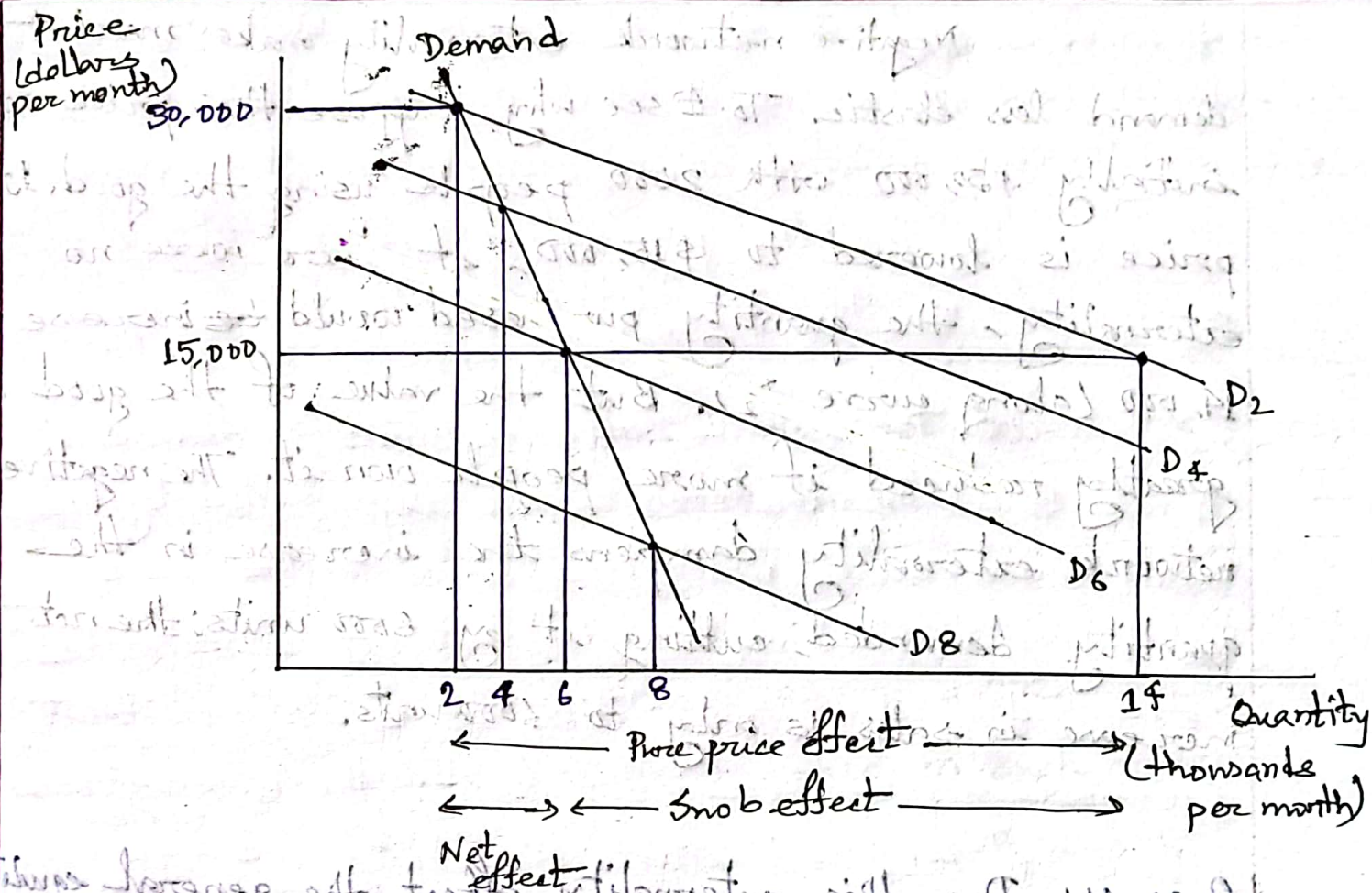
Ultimately, consumers will get a good sense of how many people have in fact purchased a product, based on its price. In figure, we see that if the price were \$30, then 40,000 people would buy the product, & demand curve is D_{40} . If the price were \$20, 80,000 people would buy the product and demand curve is D_{80} . The market demand curve is therefore found by joining the points on D_{20} , D_{40} , D_{60} , D_{80} and D_{100} that correspond to the quantities 20,000,

40,000, 60,000, 80,000 and 100,000.

Compared with the curves D_{20} , etc., the market demand curve is relatively elastic. Now consider the effect of a drop in the price from \$30 to \$20, with a demand curve of D_{40} . If there were no externality, the quantity demanded would increase from 40,000 to only 48,000. But as more people buy the product, the positive network externality increases the quantity demanded further, to 80,000. Thus, positive network externality, bandwagon effect increases the response of demand to price changes - i.e., makes the demand more elastic.

Snob Effect:-

Snob effect is a negative network externality in which a consumer wishes to own an exclusive or unique good. The quantity demanded of a "snob good" is higher the fewer people who own it. Rare works of art, specially designed sports cars and made-to-order clothing are snob goods. They are exclusivity resulting from the fact that few other people own one like it.



In the figure, we assume that the product in question is a snob good, so people value exclusively. Here, D_2 is the demand curve that would apply if consumer believed that only 2000 people used the good. If they believe that 4000 people used it, it would be less exclusive, so its value decreases and demand curve will lower, D_4 . If they believe 6000 people used it, the demand curve would be D_6 and so on. Thus the market demand curve is ~~line~~ found by joining D_2, D_4, D_6 etc, where quantities are 20,000, 40,000, 60,000 etc.

Negative network externality makes market demand less elastic. To see why, suppose the price was initially \$30,000 with 2000 people using the good. If price is lowered to \$15,000, if there were no externality, the quantity purchased would increase to 14,000 (along curve D_2). But the value of the good is greatly reduced if more people own it. The negative network externality dampens the increase in the quantity demanded, cutting it by 8000 units; the net increase in sales is only to 6000 units.

Ques-11:- Does this externality affect the general equilibrium solution in a $2 \times 2 \times 2$ model?

Answer:-

Suppose there are two consumers in the economy, A and B, whose preferences are represented by ordinal indifference curves, which are convex to the origin, exhibiting diminishing marginal rate of substitution between the two commodities. It is assumed that consumer choices are independent: the consumption patterns of A do not affect B's utility, and vice versa. Bandwagon, snob, Veblen-esque and other

external effects are ruled out. Finally, it is assumed that the consumers are sovereign, in the sense that their choice is not influenced by advertising or other activities of the firms.

So, we can say that, the consumption externality (bandwagon effect, snob effect) does not affect the general equilibrium solution in a $2 \times 2 \times 2$ model.

Ques-16:- What are the assumptions of a $2 \times 2 \times 2$ model? How does general equilibrium is reached in such model?

Answer:-

The Assumptions of the $2 \times 2 \times 2$ Model:-

1. There are two factors of production, labor (L) and capital (K), whose quantities are given exogenously. These factors are homogenous and perfectly divisible.

2. Only two commodities are produced, X and Y . Technology is given. The production functions of the two commodities are represented by two isoquant maps, with usual properties. The isoquants are smooth and convex to the origin, implying diminishing marginal rate of factor (technical) substitution along any isoquant. Each

production function exhibits constant returns to scale. Finally, it is assumed that two production functions are independent; there are no external economies or diseconomies for the production activity of one product arising from the production of the other.

3. There are two consumers in the economy, A and B whose preferences are represented by ordinal indifference curves, which are convex to the origin, exhibiting diminishing marginal rate of substitution between the two commodities. It is assumed that consumer choices are independent; the consumption patterns of A do not affect B's utility, and vice versa. Bandwagon, Snob, Veblen-esque, and other external effects are ruled out. Finally, it is assumed that the consumers are sovereign, in the sense that their choice is not influenced by advertising or other activities of the firms.

4. The goal of each consumer is the maximization of his own satisfaction (utility), subject to his income constraint.

5. The goal of each firm is profit maximization, subject to the technological constraint of the production

function. $P^M, P^M - P^M$ test work - 11-2018

7. There is full employment of the factors of production, and all incomes are received by their owners (A and B) are spent.

8. The factors of production are owned by the consumers.

8. There is perfect competition in the commodity and factor markets. Consumers and firms pursue their goals faced by the same set of prices (P_x, P_y, w, r).

In this model a general equilibrium is reached when the four markets (two commodity markets and two factor markets) are cleared at a set of equilibrium prices (P_x, P_y, w, r) and each participant economic agent (two firms and two consumers) is simultaneously in equilibrium.

$$P^M - P^M = P^M$$

$$P^M - P^M = P^M$$

Ques-17:- Show that $MRPT_{x,y} = MC_x / MC_y$

Answer:-

(i) By definition,

$$MC_x = \frac{d(TC_x)}{dX}$$

$$\text{and } MC_y = \frac{d(TC_y)}{dY}$$

(ii) Dividing these expressions we obtain

$$\frac{MC_x}{MC_y} = \frac{d(TC_x)}{d(TC_y)} \cdot \frac{dY}{dX} \quad (2a)$$

(iii) But

$$d(TC_x) = w \cdot (dL_x) + r \cdot (dK_x)$$

$$\text{and } d(TC_y) = w \cdot (dL_y) + r \cdot (dK_y)$$

$$\text{so that, } \frac{d(TC_x)}{d(TC_y)} = \frac{w \cdot (dL_x) + r \cdot (dK_x)}{w \cdot (dL_y) + r \cdot (dK_y)} \quad (3a)$$

(iv) In order to remain on the PPC the factors released from the decrease in commodity Y must be equal to the factors absorbed by the increase in the production of X, that is

$$dL_x = -dL_y$$

$$\text{and } dK_x = -dK_y$$

----- (4a)

Substituting (4) in (3) we find

$$\frac{d(TC_x)}{d(TC_y)} = \frac{w(-dL_y) + r(-dK_y)}{w(dL_y) + r(dK_y)} = -1 \quad (5a)$$

(v) Finally, substituting (5) in (2), we obtain

$$\frac{MC_x}{MC_y} = (-1) \frac{dY}{dX} = - \frac{dY}{dX} = (\text{slope of PPC}) = MRPT_{x,y}$$

In perfect competition the profit-maximizing producer equates the price of the commodity produced to the long-run marginal cost of production:

$$MC_x = P_x \quad \text{and} \quad MC_y = P_y$$

Therefore the slope of the production possibility curve is also equal to the ratio of the prices at which X and Y will be supplied by perfectly competitive industries:

$$MRPT_{x,y} = \frac{MC_x}{MC_y} = \frac{P_x}{P_y}$$

(shown)

Que-15:- Present a 2x2x2 Walrasian system. Does the general equilibrium solution exist in such model? Explain.

Answer:-

The Walrasian System:-

The most ambitious general equilibrium model was developed by the French economist Leon Walras. He argued that all prices and quantities in all markets are determined simultaneously through their interaction with one another. He used a system of simultaneous equations to explain the interaction of buyers and sellers in all markets.

In a general equilibrium system of the Walrasian type there are as many markets as there are commodities and factors of production.

For each market there are three types of functions: demand functions, supply functions and a 'clearing-the-market' equation.

In a commodity market the number of demand functions is equal to the number of consumers,

and the number of supply functions is equal to the number of firms which produce the commodity.

In each factor market the number of demand functions ^{is} equal to the number of firms multiplied by the number of commodities they produce. The number of supply function is equal to the number of consumers who own the factors of production.

A necessary (but not sufficient) condition for the existence of a general equilibrium is that there must be in the system as many independent equations as the number of unknowns.

For example, assume that an economy consists of two consumers, A and B, who own two factors of production, K and L. These factors are used by two firms to produce two commodities, X and Y. (This is the simple $2 \times 2 \times 2$ general equilibrium model). It is assumed that each firm produces one commodity, and each consumer buys some quantity of both. It is also assumed that both consumers own some quantity of both factors. In this simple model we have the following 'unknowns':

quantities demanded of X and Y by consumers	$2 \times 2 = 4$
quantities supplied of K and L by consumers	$2 \times 2 = 4$
quantities demanded of K and L by firms	$2 \times 2 = 4$
quantities of Y and X supplied by firms	2
prices of commodities Y and X	2
prices of factors, K and L	2

Total number of unknowns' 18

To find these unknowns we have the following number of equations:

demand functions of consumers	$2 \times 2 = 4$
supply functions of factors	$2 \times 2 = 4$
demand functions of factors	$2 \times 2 = 4$
supply functions of commodities	2
clearing-the-market of commodities	2
clearing-the-market of factors	2

Total number of equations 18

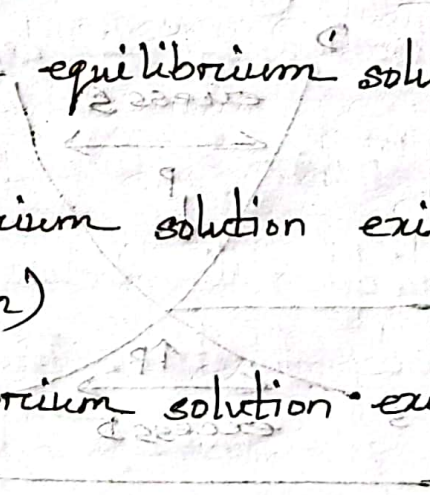
Since the number of equations is equal to the number of unknowns, one should think that a general equilibrium solution exists. Unfortunately, the equality of numbers of equations and unknowns is neither a sufficient nor a necessary condition for the existence

of a solution. So even if there is equality of equations and unknowns, there is no guarantee that a general equilibrium solution exists. The proof of the existence of a general equilibrium solution is difficult. Leon Walras was never able to prove the ~~general~~ existence of a general equilibrium.

Ques-09:- Graphically explain the problems of existence, uniqueness and stability in connection to general equilibrium

Answer:-

(1) Three problems arise in connection with a general equilibrium problem:-

- 
- (i) Does a general equilibrium solution exist? (Existence problem)
 - (ii) If an equilibrium solution exists, is it unique? (Uniqueness problem)
 - (iii) If an equilibrium solution exists, is it stable? (Stability problem)

These problems can best be illustrated with the partial-equilibrium example of a demand-supply model. Assume that a commodity is sold in a perfectly

competitive market, so that from the utility-maximizing behaviour of individual consumers there is a market demand function, and from the profit-maximizing behaviour of firms there is a market supply function. The price at which $Q_D = Q_S$ is the equilibrium price. At such a price there is neither excess demand nor excess supply.

The equilibrium is stable if the demand function cuts the supply function from above. In this case an excess demand drives ~~up~~ price up, while an excess supply drives the price down (Fig-1)

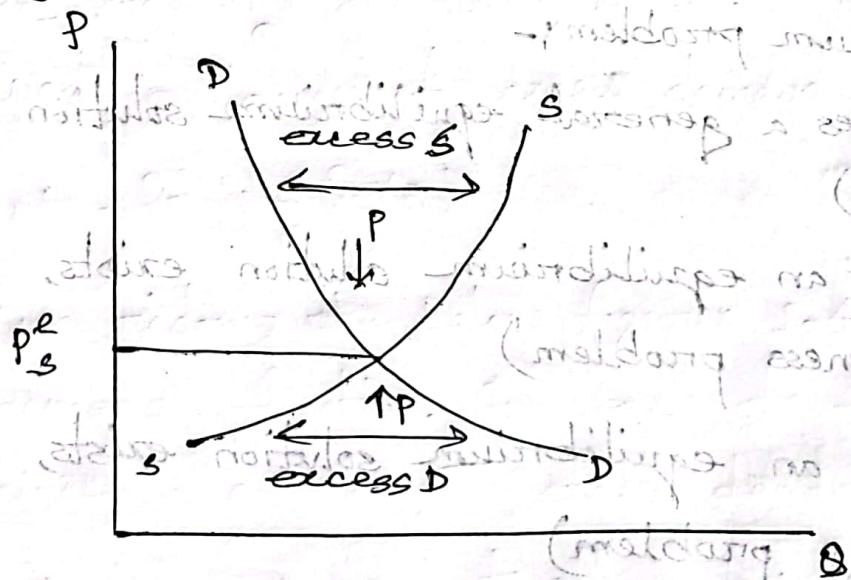


Fig-1: Unique, stable equilibrium

The equilibrium is unstable if the demand function cuts the supply function from below. In this

case, an excess demand drives the price down, and an excess supply drives the price up (figure-2)

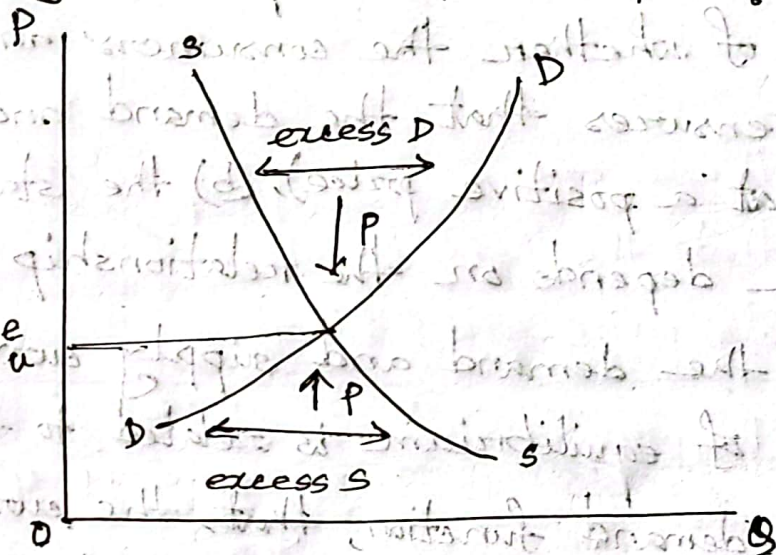


Fig-2: Unique, unstable equilibrium

In fig-3, we depict the case of multiple equilibria. It is obvious that at P_1^e there is a stable equilibrium, while at P_2^e the equilibrium is unstable. Finally, in fig-4 an equilibrium (at a positive price) does not exist.

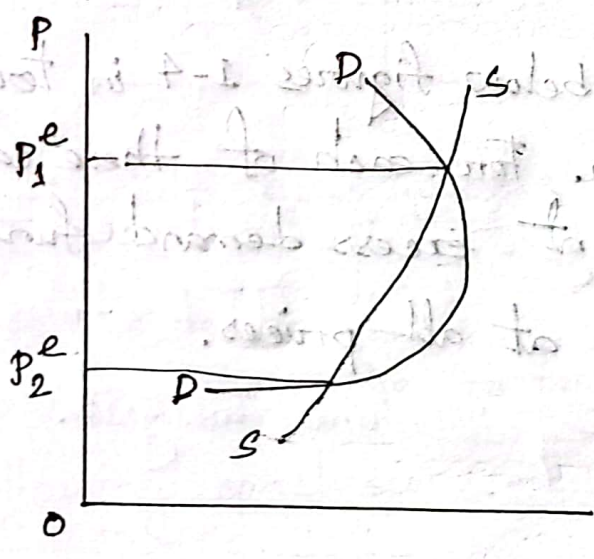


Fig-3: Multiple equilibria

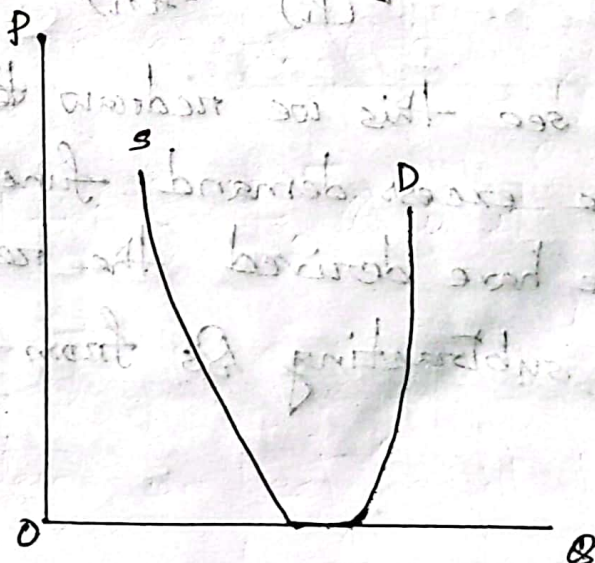


Fig-4:- No equilibrium exists

It should be clear from the above discussion that (a) the existence of an equilibrium is related to the problem of whether the consumers and producers behaviour ensures that the demand and supply curves intersect (at a positive price); (b) the stability of equilibrium depends on the relationship between the slopes of the demand and supply curves; (c) the uniqueness of equilibrium is related to the slope of the excess demand function, that, the curve which shows the difference between D_D and D_S at any one price.

In fact the three basic questions related to the existence, stability and uniqueness of an equilibrium can be expressed in terms of excess demand function:

$$E(P_i) = D_D(P_i) - D_S(P_i)$$

To see this we redraw ~~the~~ below figures 1-4 in terms of the excess demand function. For each of these cases we have derived the relevant excess demand function by subtracting D_S from D_D at all prices.



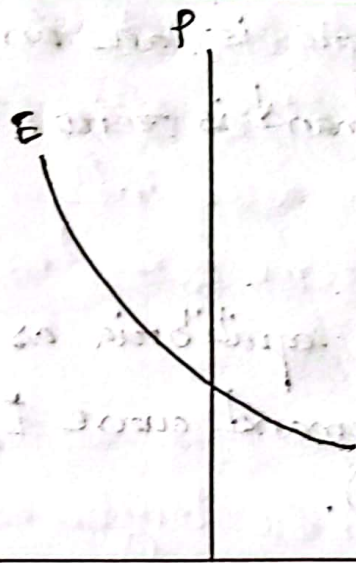


Fig-5: -stable equilibrium: slope of $E(P) < 0$

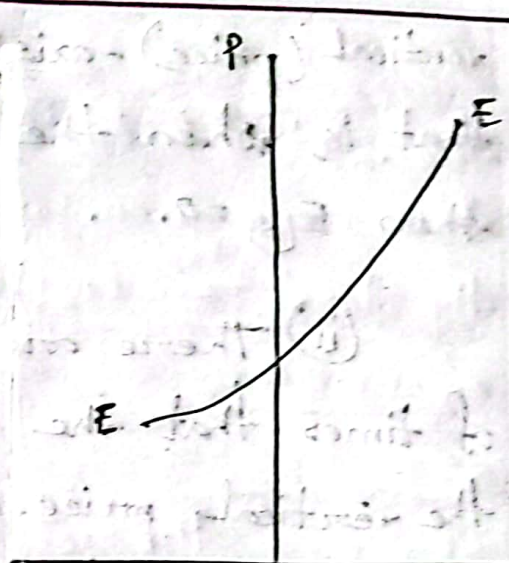


Fig-6: Unstable equilibrium: slope of $E(P) > 0$

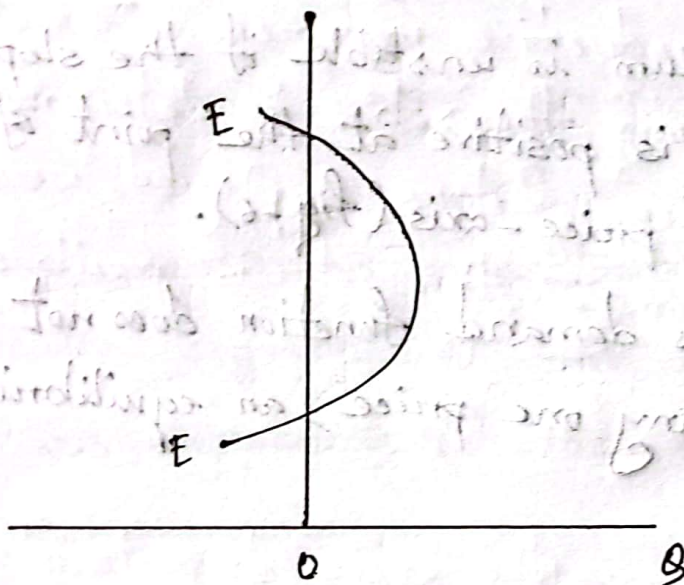


Fig-7: Multiple equilibria

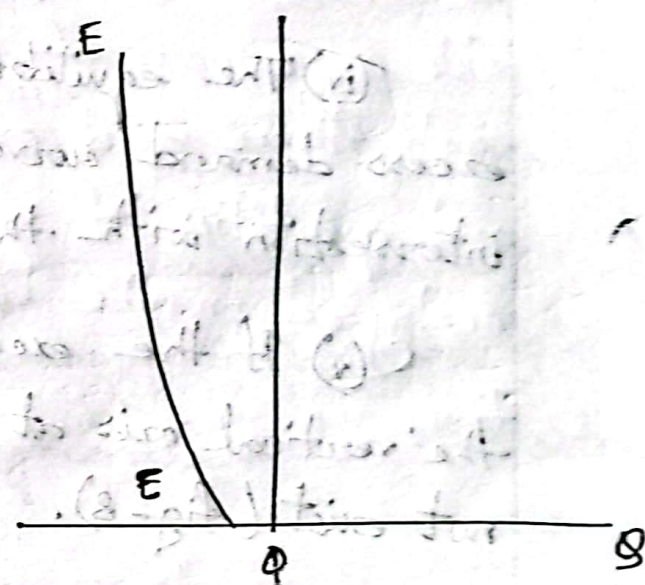


Fig-8: No equilibrium exists

From the redrawn diagrams, we can draw the following conclusions:-

- (i) The excess demand function, $E(P)$, intersects the

vertical (price) - axis where there is an equilibrium that is, when the excess demand is zero. If $D_p = S_p$, then $E(p) = 0$.

(ii) There are as many equilibria as the number of times that the excess demand curve $E(p)$ intersects the vertical price-axis (fig-7).

(iii) The equilibrium is stable if the slope of the excess demand curve is negative at the point of its intersection with the price-axis (fig-5).

(iv) The equilibrium is unstable if the slope of the excess demand curve is positive at the point of its intersection with the price-axis (fig-6).

(v) If the excess demand function does not intersect the vertical axis at any one price, an equilibrium does not exist (fig-8).

The above analysis of the existence, stability and uniqueness in terms of excess demand functions can be extended to general equilibrium analysis.

Ques-14:- Define general equilibrium? In what respect it differ from the partial equilibrium? Examine with example.

Answer:-

General Equilibrium:-

A general equilibrium is defined as a state in which all markets and all-decision making units are in simultaneous equilibrium. A general equilibrium is said to exist if each market is cleared at a positive price with each consumer maximizing satisfaction and each firm maximizing profit.

Different from Partial Equilibrium:-

A partial equilibrium approach, ~~concerns~~ concentrating on decisions in a particular segment of the economy in isolation of what was happening in other segments, under the ceteris paribus assumption.

The utility-maximizing behaviour of a household under the assumption that its income was given, although income depends on the amount of labour and other factors of production that the consumer owns and on their prices. Likewise the production decision of a

firm depends on the assumption that factor prices, the state of technology and the prices of commodities were given. But they are influenced by the level of employment, income and tastes of consumers.

Product markets, where buyers and sellers interacted with each other and among themselves to determine prices and levels of outputs of various commodities; factor markets, where firms and households as owners of productive resources interacted with each other and among themselves to determine prices and quantities of various factors employed, were also analysed on the basis of the ceteris paribus assumption.

In summary, the basic characteristics of a partial equilibrium approach is the determination of the price and quantity in each market by demand and supply curves drawn on ceteris paribus clause.

However, a fundamental feature of any

economic system is the interdependence among its constituent parts. The markets of all commodities and all productive factors are interrelated, and the prices in all markets are simultaneously determined. For example, consumers' demands for various goods and services depend on their tastes and incomes. $\rightarrow$ In turn, consumers' incomes depend on the demand and supply of the various inputs. $\rightarrow$ The demand for factors by firms depends not only on the state of technology but also on the demands for the final goods they produce. $\rightarrow$ The demands for these goods depend on consumers' income, which, as we saw, depends on the demand for ^{the} factors of production. This circular interdependence of the activity within an economic system can be illustrated with a simple economy composed of two sectors, a consumer sector, which includes households, and a business sector, which includes firms. It is assumed that: (a) all production takes place in the business sector; (b) all factors of production are owned by the households; (c) all factors are fully employed; (d) all incomes are spent.

The economic activity in the system takes the form of two flows between the consumer sector

and the business sector: a real flow and a monetary flow.

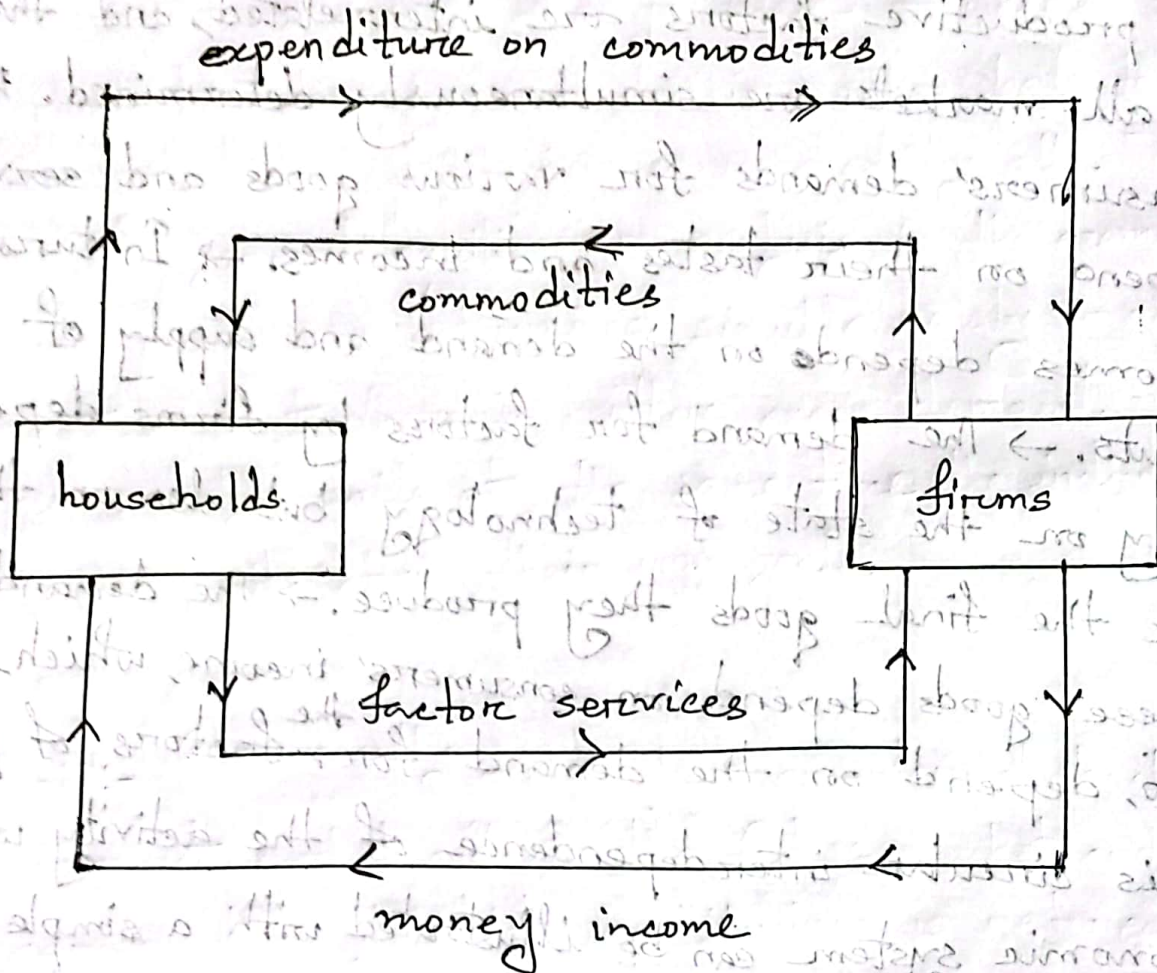


Fig:- Circular flow in a two-sector economy

The real flow is the exchange of goods for the services of factors of production: firms produce and offer final goods to the household sector and consumers offer to firms the services of factors which they own.

The monetary flow is the real flow expressed

in monetary terms. The consumers receive income payments from the firms for offering their factor services. These incomes are spent by consumers for the acquisition of the finished goods produced by the business sector.

The real flow and the monetary flow move opposite directions. They are linked by the prices of goods and factor services. The economic system is in equilibrium when a set of prices is attained at which the magnitude of the income flow from firms to households is equal to the magnitude of the money expenditure flow from households to firms.

The ~~an~~ interdependence of markets concealed by the partial equilibrium approach. Markets consist of buyers and sellers. Thus an economic system consists of millions of economic decision-making units who are motivated by self-interest. Everyone is maximizing something. The consumer maximises satisfaction, the firm maximizes profit, a worker determines his supply of labor by maximizing satisfaction derived from work-leisure opportunities.

The problem is to determine whether the independent, self-interest motivated behaviour of economic

decision makers is consistent with each ~~other~~ individual agent's attaining ~~behaviour~~ equilibrium. General equilibrium deals with the problem of whether the independent action by each decision maker leads to a position in which equilibrium is reached by all. The scope of general equilibrium analysis is the examination of how prices are determined simultaneously in all markets, so that there is neither excess demand, nor excess supply, while at the same time the individual economic units attain their own goals.

Ques-12:- Graphically explain how an external shock in the product market affects the factor market and the mechanism through which equilibrium is restored in both market.

Answer:-

A Graphical Illustration of the path to General Equilibrium:-

Here we attempt to show a simple economy with perfectly competitive production and factor markets will have an inescapable tendency toward a general equilibrium solution.

Assumptions:-

(i) Two substitute commodities are produced, X and Y, by two

perfectly competitive industries (markets).

(ii) There are two factors, capital K and labor L , whose markets are perfectly competitive. The quantities of these factors are given (fixed supply).

(iii) The production functions are continuous, with diminishing marginal rate of factor substitution and decreasing returns to scale.

(iv) The industry producing X is less capital intensive than the industry producing Y . The K/L ratio is smaller in industry X .

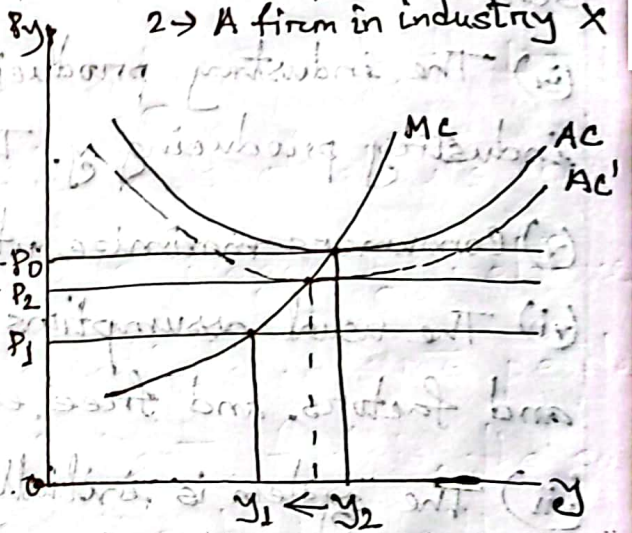
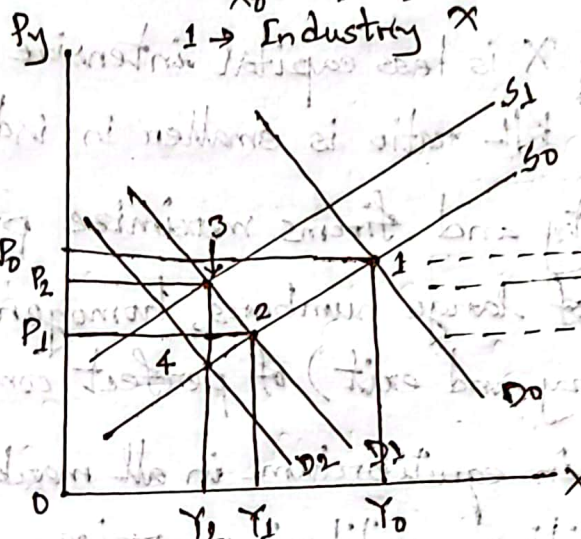
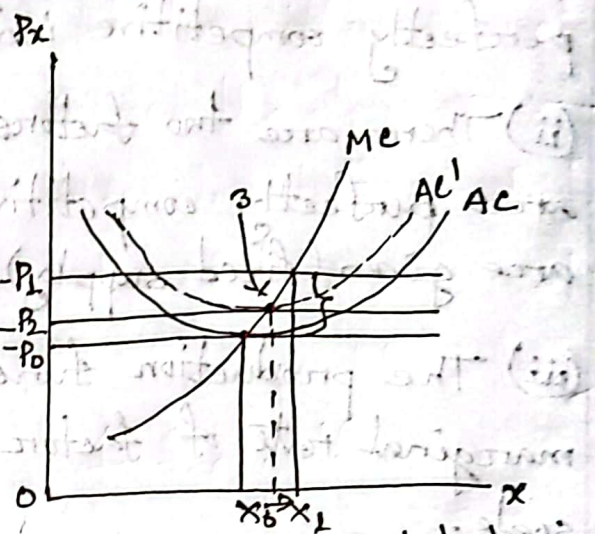
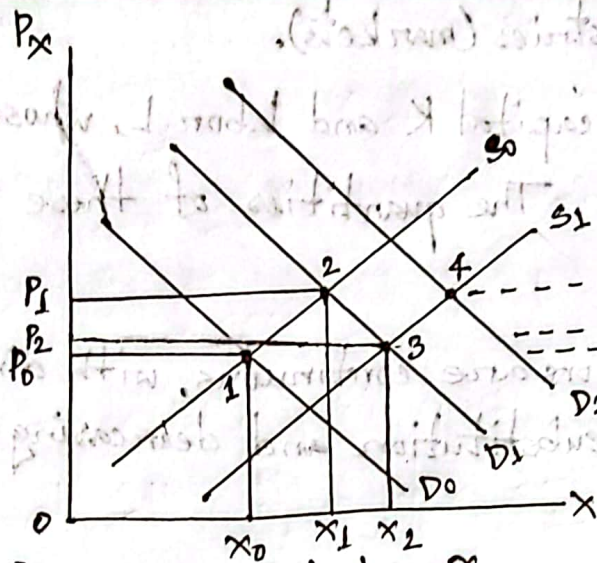
(v) Consumers maximize utility and firms maximize profit.

(vi) The usual assumptions (of large numbers, homogenous products and factors, and free entry and exit) of perfect competition hold.

(vii) The system is initially in equilibrium in all markets demand is equal to supply at a positive equilibrium price.

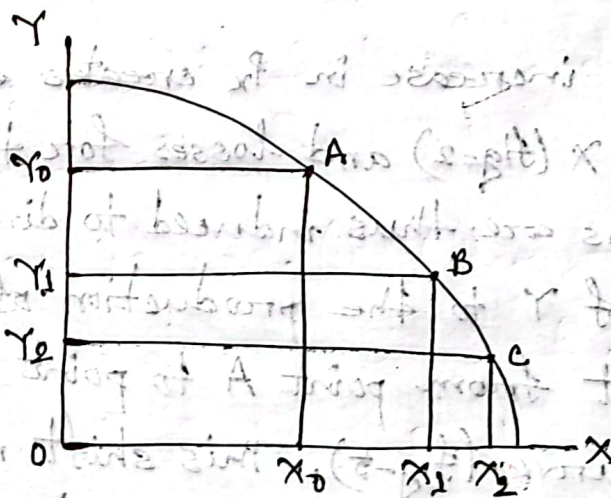
The increase in P_X creates excess profits for the producers of X (Fig-2) and losses for the producers of Y (Fig-4). Firms are thus induced to divert resources from the production of Y to the production of X . This is shown by the movement from point A to point B on the production possibility curve (Fig-5). This shift reflects the effect of the change in consumers' tastes on the decisions of firms.

The increase in P_X induces the producers of X to increase their quantity in order to maximize their profit,



3 → Industry Y

4 → A firm in industry Y



5- Production possibility curve

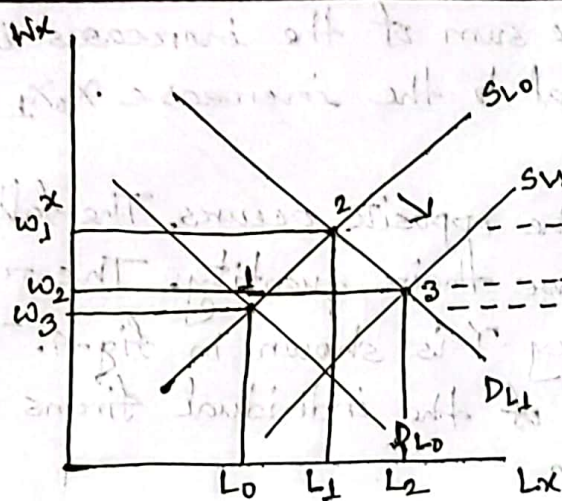
given the increase in marginal revenue (shown in fig-2). Since every firm in the X industry faces the same price the output of each firm increases (each firm produces on the

raising part of its AC). The sum of the increases in outputs of existing firms is equal to the increase X_0X_1 in fig-1.

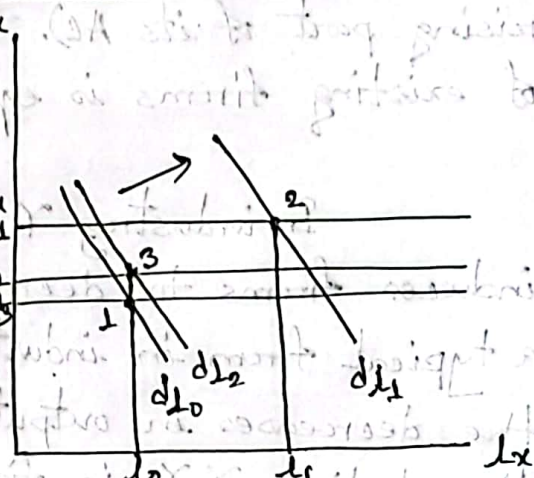
In industry Y the opposite occurs. The fall in P_Y induces firms to decrease their quantity. The reaction of a typical firm in industry Y is shown in fig-4. The sum of the decreases in output of the individual firms is equal to the decline Y_0Y_1 in fig-3.

In the long run excess profits attract entry in industry X and induce exit of firms from industry Y. Entry and exit affect the demand for factors of production. The markets for labor and capital used in the production of X are shown in fig-6 and 8.

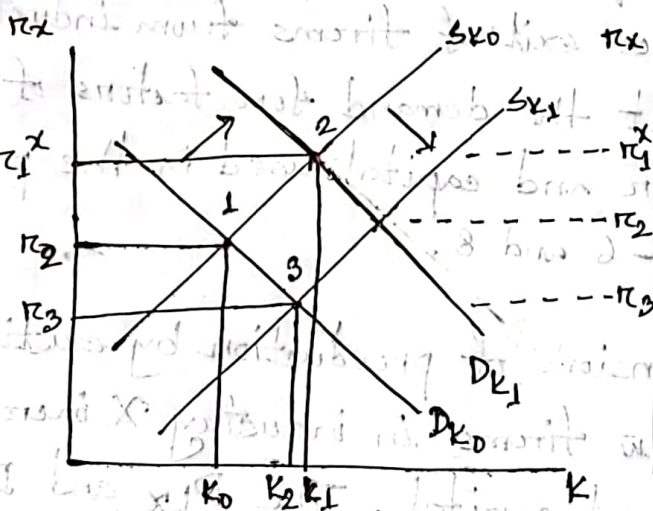
The expansion of production by existing firms and the entry of new firms in industry X increases the demand for labor and capital. The DL_X and DK_X curves shift outwards, and w and r_X rise. Employment of these factors rises in industry X (by L_0L_1 and K_0K_1 respectively in fig-6 and 8). The demand for L and K by a single firm in industry X is shown in fig-7 and 9. The individual firm can buy any quantity of L and K at the prevailing market price. At the increased price w_1^X the firm will demand L_1 of labor (fig-6), and at the increased price r_1^X the firm will demand K_1 of capital (fig-8).



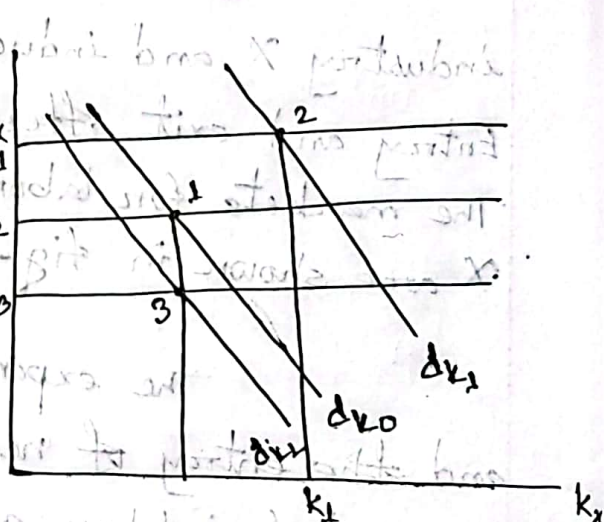
6 → labor market for industry X



7 → Demand for labor by a firm in industry X



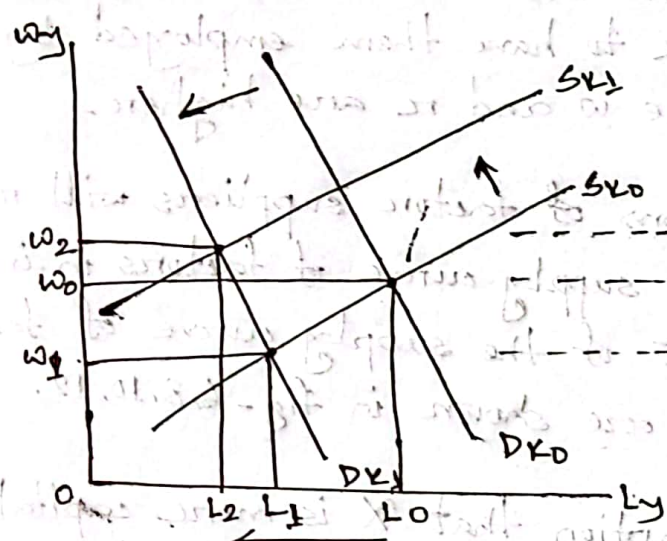
8 → Market for capital in industry X



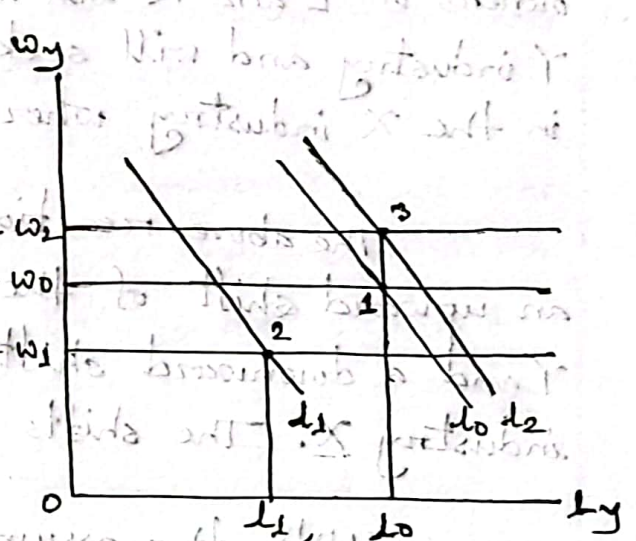
9 → Demand for capital by a firm in industry X

The situation in the labor and capital markets for the industry Y will be the reverse. The surviving firms in this industry will reduce their demand for both factors, while the exit of firms in the long run will further reduce the demand for these inputs. The situation for the factor markets in industry Y is shown in fig-10 and 12. The case of a single firm in industry Y is depicted in fig-11

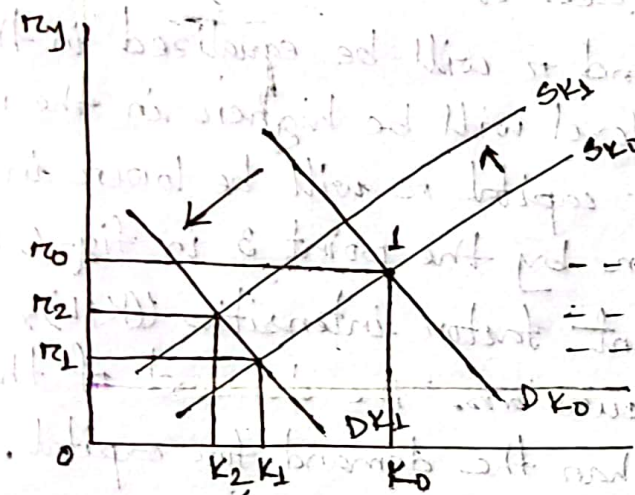
and 13.



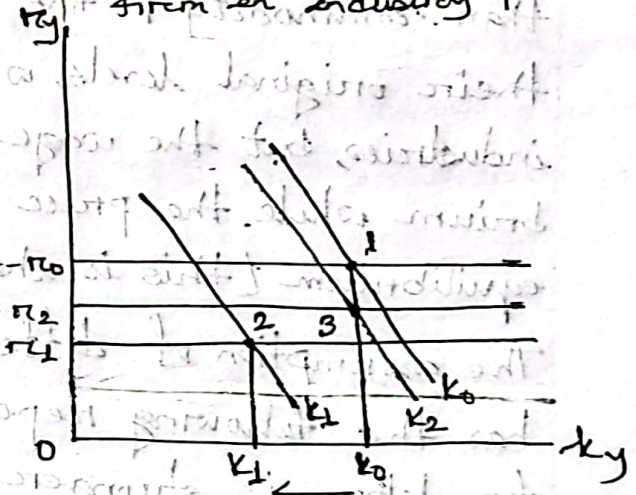
10 → Labor market for industry Y



11 → Demand for labor by a firm in industry Y



12 → Market for capital for industry Y



13 → Demand for capital by a firm in industry Y

The above change in the demands for factors in the two industries leads to disequilibrium, because the prices of L and K have risen in the factor markets for industry X , while w and r have fallen in the factor markets of industry Y . However, in perfect factor markets the disequilibrium will be self-correcting, since in the long run there is perfect mobility of factors between the different markets. Thus the

owners of L and K will withdraw their services from the Y industry and will seek to have them employed by firms in the X industry where w and r are higher.

The above reactions of factor suppliers will result in an upward shift of the supply curve of factors in industry Y and a downward shift of the supply curve of factors in industry X. The shifts are shown in fig- 6, 8, 10, 12.

With the assumption that X is more capital intensive than commodity Y the prices of factors will not return to their original levels, w and r will be equalized in the two industries, but the wage level will be higher in the new equilibrium while the price of capital r will be lower in the final equilibrium (this is shown by the point 3 in fig- 6, 8, 10, 12).

The assumption of different factor intensities $(K/L)_X < (K/L)_Y$ has the following repercussions. The demand of the X industry for labor is stronger than the demand for capital.

The release of labor by industry Y is smaller than the rate required by X, while the release in capital is larger than the increased need for this factor by industry X. Thus overall the demand for labor will be higher than initially and w will rise. The opposite will occur in the market for capital, where new equilibrium r will be lower than in the initial situation.

As a result of the new factor prices, the individual firms in each industry will adjust their demands for labor



and capital. In figures 7, 9, 11, 13 the individual firms are in equilibrium at the points denoted by (3).

The entry of firms in industry X shifts the supply curve downwards to S_1 (fig-1). The final equilibrium price of X is lower than in the short run, but higher than in the original equilibrium. The X industry is an increasing cost industry.

The exit of firms from industry Y shifts the supply of this industry upwards (fig-3). The final equilibrium price P_2 is higher than the short-run price P_1 but lower than the initial equilibrium level P_0 . Industry Y is also an increasing cost industry.

Given the new product prices the individual firms will adjust their outputs. Firms in industry X will be producing an output lower than in the short run, but higher than in the initial equilibrium (fig-2).

Firms in industry Y will be producing an output higher than in the short run, but lower than in the initial situation (fig-4). The changes in w and r cause the LAC of firms in X to shift upwards, and LAC of firms in Y to shift downwards. Thus in the long run the firms earn just normal profits.

If we accept the fact that wage earners have a higher propensity to consume than other factor owners, there

will be a further increase in the demand for X , shifting the demand curve to D_2 in fig-1. From this point we can start tracing the additional changes in all other markets, as previously.

The above discussion is not a formal proof of the existence or stability of a general equilibrium. However, the assumption of perfect competition was adequate to enable us to see that the shifts in the supply and demand curves are not random, but precisely interrelated. The curves will continue to shift so long as disequilibria exist. When an equilibrium is attained simultaneously in all markets and among all economic decision-makers within the markets, we say that a general equilibrium exists throughout the entire economy, and the movement of curves will cease until another exogenous force disturbs the system.

Ques-13:- Explain the production equilibrium/consumption equilibrium/simultaneous equilibrium in a simple general equilibrium model through Edgeworth box diagram.

Answer:-

Three static properties are observed in a general equilibrium solution, reached with a free competitive market mechanism:-

(a) Efficient allocation of resources among firms (equilibrium of production).

(b) Efficient distribution of the commodities produced between the two consumers (equilibrium of consumption).

(c) Efficient combination of products (simultaneous equilibrium of production and consumption).

These properties are called the marginal conditions of Pareto optimality. A situation is Pareto optimal if it is impossible to make anyone better-off without making someone worse-off. Now we'll discuss briefly the three optimality properties that are observed in a general equilibrium state.

(a) Equilibrium of production (efficiency in factor substitution):-

Equilibrium of production requires the determination of the efficient distribution of the available productive factors among the existing firms (efficiency in factor substitution).

We know that, the firm is in equilibrium if it chooses the factor combination which minimizes its cost. Thus the equilibrium of the firm requires that

$$[\text{slope of isoquant}] = [\text{slope of isocost}]$$

$$\text{or, } MRTS_{L,K} = \frac{w}{r}$$

where w and r are the factor prices prevailing in the market and $MRTS$ is the marginal rate of technical substitution between the factors.

The joint equilibrium of production of the two firms in our simple model can be derived by the use of the

Edgeworth box of production. On the axes of this construct we measure the given quantities of the factors of production, K and L (Fig-1). The isoquants of commodity X are plotted with origin the south-west corner and the isoquants of Y are plotted with origin the north-east corner. The locus of points of tangency of the X and Y isoquants is called the Edgeworth contract curve of production.

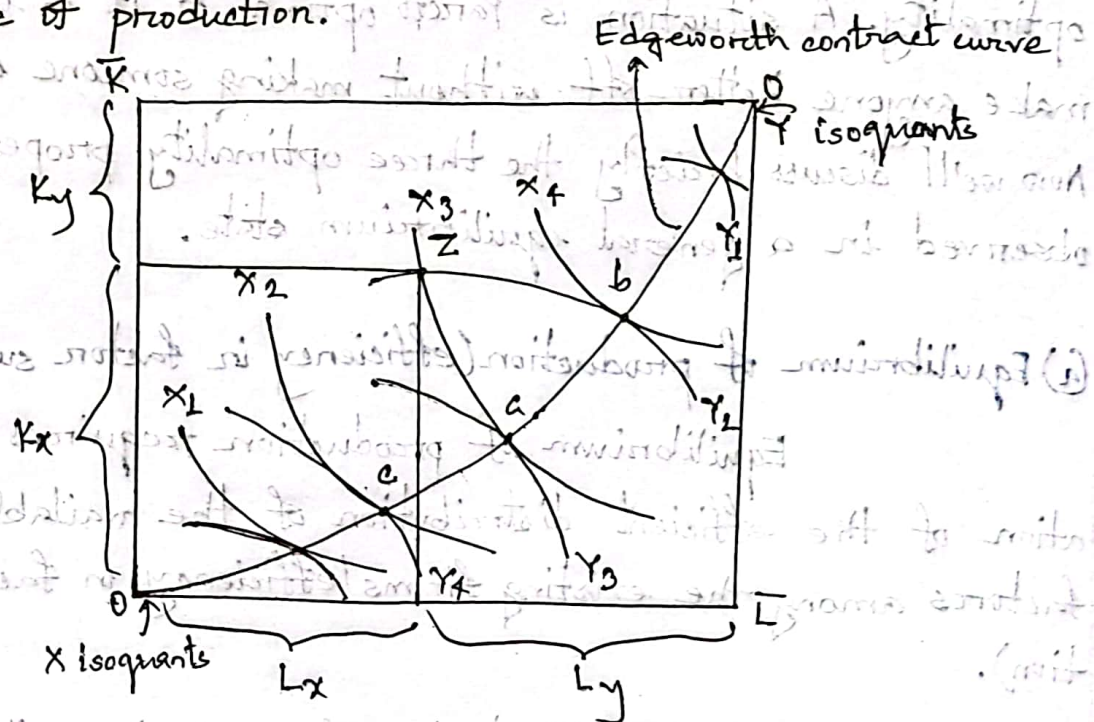


Fig-1: Edgeworth box of production

Each point of the Edgeworth box shows a specific allocation of K and L in the production of commodities X and Y . The amounts of Y and X produced and the amounts of capital and labor allocated to the production of Y and X are shown. For example point Z shows that:

- X_3 is the quantity produced of commodity X .
- Y_2 is the quantity produced of commodity Y .
- K_x is the amount of capital allocated to the production of X_3 .

• K_y is the amount of capital allocated to the production of Y_2 .

• L_x is the amount of labor allocated to the production of X_3 .

• L_y is the amount of labor allocated to the production of Y_2 .

However, not all points of the Edgeworth box represent efficient allocations of the available resources. An allocation of inputs is efficient if the produced combination of X and Y is such that it is impossible to increase the production of one commodity without decreasing the quantity of other. From fig-1 we see that efficient production takes place on the Edgeworth contract curve. It is impossible to move to a point off this curve without reducing the quantity of at least one commodity. Point Z is thus a point of inefficient production.

Since the edgeworth contract curve of production is the locus of tangencies of the X and Y isoquants, at each of its points the slopes of the isoquants are equal:

$$[\text{slope of } X \text{ isoquant}] = [\text{slope of } Y \text{ isoquant}]$$

or,

$$MRTS_{L,K}^X = MRTS_{L,K}^Y$$

~~Ques~~ In our simple general equilibrium model the firms, being profit maximizers in competitive markets, will be in equilibrium only if they produce somewhere on the Edgeworth contract curve. The factor prices facing the producers are

the same, and their profit maximization requires that each firm equates its $MRTS_{L,K}$ with the ratio of factor prices w/r :

$$MRTS_{L,K}^x = MRTS_{L,K}^y = \frac{w}{r}$$

In summary, the general equilibrium of production occurs at a point where the $MRTS_{L,K}$ is the same for all firms, that is, at a point which satisfies the Pareto optimality criterion of efficiency in factor substitution: $MRTS_{L,K}$ is equal to the ratio of the market factor prices w/r .

(6) Equilibrium of consumption (efficiency in distribution of commodities):-

We now show how each consumer, faced with the market prices P_x and P_y , reaches equilibrium, that is, maximizes his satisfaction. We know that the consumer maximizes his utility by equating the marginal rate of substitution of the two commodities (slope of ~~the~~ his indifference curves) to the price ratio of the commodities. Thus the condition for consumer equilibrium is

$$MRS_{x,y} = \frac{P_x}{P_y}$$

Since both consumers in perfectly competitive markets are faced with the same prices the condition for joint or general equilibrium of both consumers is

$$MRS_{x,y}^A = MRS_{x,y}^B = \frac{P_x}{P_y}$$

This general equilibrium of consumption for the product mix Y^e, X^e is shown in fig-2:

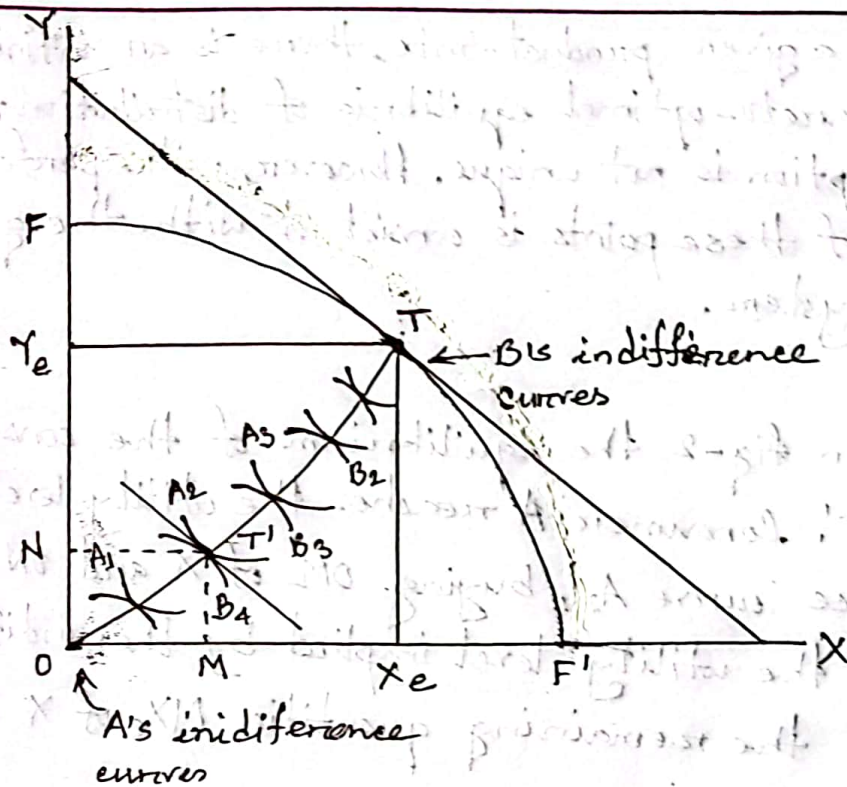


Fig-2

In fig-2, we construct an Edgeworth box for consumption with the precise dimensions Y^e and X^e by dropping from point T lines parallel to the commodity axes. We next plot the indifference curves of consumer A with origin the south-west corner, and the indifference curves of B with origin the north-east corner.

From fig-2 it is seen that only points of tangency of the indifference curves of the two consumers represent Pareto-efficient distributions. The locus of these points ~~are~~ is called the Edgeworth contract curve of consumption. It should be clear that at each point of this curve the following equilibrium condition is satisfied

$$MRS_{x,y}^A = MRS_{x,y}^B$$

Thus for a given product-mix, there is an infinite number of possible Pareto-optimal equilibria of distribution; the equilibrium of consumption is not unique. However, with perfect competition, only one of these points is consistent with the general equilibrium of the system.

In fig-2 the equilibrium of the consumers is defined by point T'. Consumer A reaches the utility level implied by the indifference curve A_2 , buying OM of X and ON of Y . Consumer B reaches the utility level implied by the indifference curve B_4 , buying the remaining quantities MX_e of X and NY_e of Y .

(c) Simultaneous equilibrium of production and consumption (efficiency in product-mix):-

The general equilibrium of the system as a whole requires the fulfillment of a third condition, namely that the marginal rate of product transformation (slope of the PPC) be equal to the marginal rate of substitution of the two commodities between the consumers

$$MRPT_{x,y} = MRS_{x,y}^A = MRS_{x,y}^B$$

In perfect competition this condition is satisfied, since

$$MRPT_{x,y} = \frac{P_x}{P_y}$$

and

$$MRS_{x,y}^A = MRS_{x,y}^B = \frac{P_x}{P_y}$$

so that

$$MRPT_{x,y} = MRS_{x,y}^A = MRS_{x,y}^B$$

This is the third condition of Pareto efficiency. It refers to the efficiency of product substitution. Since the MRPT shows the rate at which a good can be transformed into another in production, and the MRS shows the rate at which the consumers are willing to exchange one good for another, the system is not in equilibrium unless the two ratios are equal. A simple numerical example may illustrate the argument. Suppose that the $MRPT_{x,y}$ is $2Y/X$, while the $MRS_{x,y} = Y/X$. The economy can produce two units of Y by sacrificing one unit of X , while the consumers are willing to exchange one unit of X for one unit of Y . In fig-3 the inequality of the two ratios is shown by points c and d .

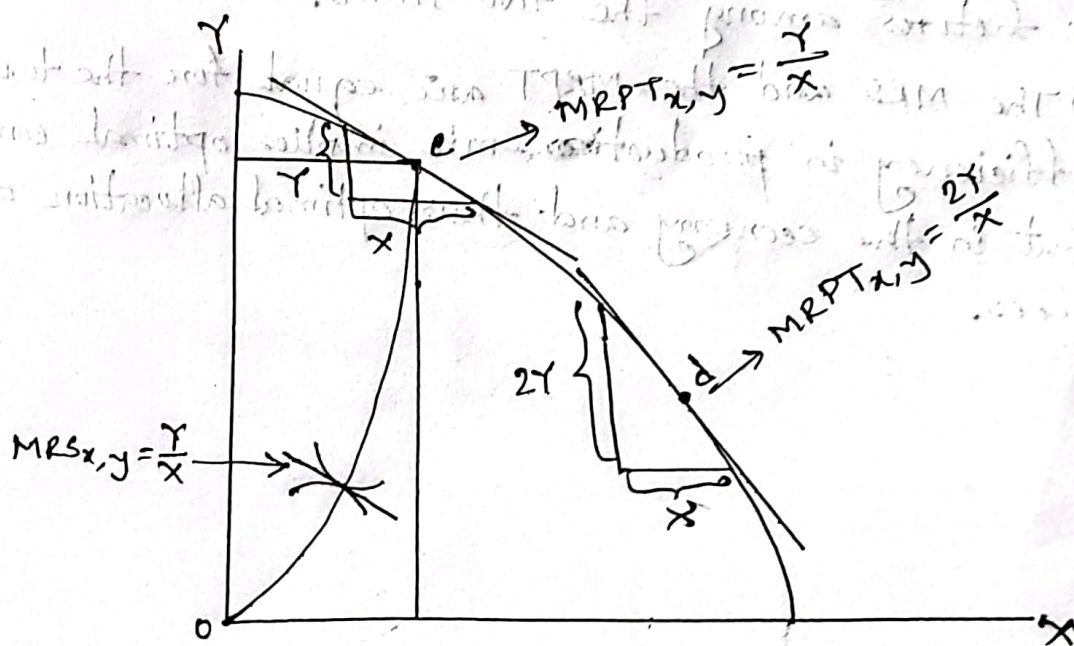


Fig-3: Disequilibrium of production and consumption

Apparently firms produce a smaller quantity of Y and a larger quantity of X relative to the preferences of the consumers. Given the assumption of consumer sovereignty, firms

must reduce X and increase the production of Y for the attainment of general equilibrium.

In summary, with perfect competition the simple two factor, two-commodity, two-consumer system has a general equilibrium solution, in which three Pareto-efficiency conditions are satisfied:-

(1) The MRS between the two goods is equal for both consumers. The efficiency in distribution implies optimal allocation of the goods among consumers.

(2) The MRTS between the two factors is equal for all firms. This efficiency in factor substitution implies optimal allocation of the factors among the two firms.

(3) The MRS and the MRPT are equal for the two goods. This efficiency in production mix implies optimal composition of output in the economy and thus optimal allocation of resources.

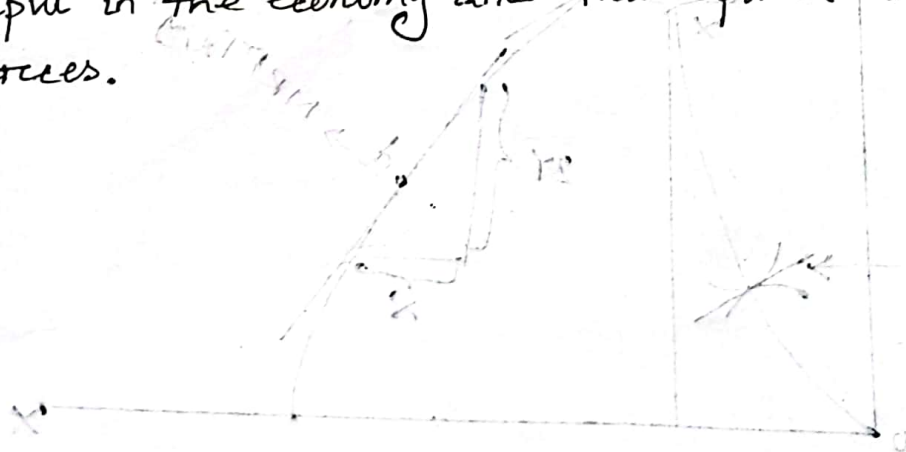


Fig. 3: Distribution of production factors